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Ryan

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(54) **STRAW HOLDER ASSEMBLY FOR A DISHWASHER APPLIANCE**

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A47L 15/42	(2006.01)
A47L 15/50	(2006.01)
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(52) **U.S. Cl.**

CPC ***A47L 15/503*** (2013.01); ***A47L 15/14*** (2013.01); ***A47L 15/4214*** (2013.01); ***A47L 15/4278*** (2013.01); ***A47L 15/507*** (2013.01); ***B08B 9/0323*** (2013.01); *B08B 2209/032* (2013.01)

(58) **Field of Classification Search**

CPC A47L 15/16

See application file for complete search history.

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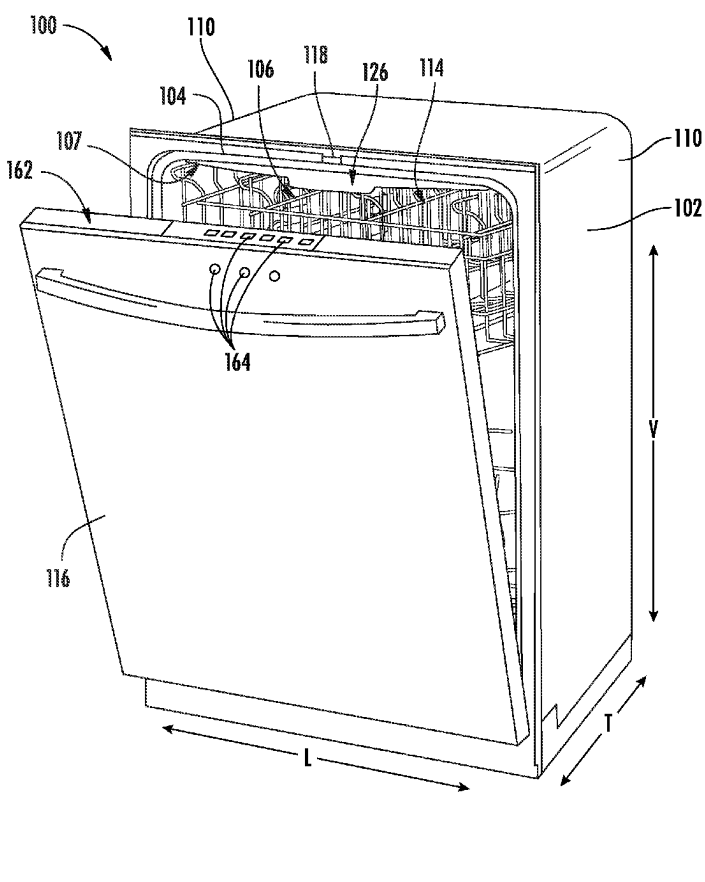
(57) **ABSTRACT**

A dishwasher appliance and a wash assembly for tubular articles in a dishwasher includes a base forming a conduit, a nozzle fluidly coupled to the conduit and configured to fit within a reusable straw such that the nozzle provides a dedicated flow of wash fluid to the inside of the straw. The wash assembly includes a retainer system to removably secure the straw to the wash assembly. The retainer system may include a resilient clip secured to the nozzle and deformable to accept and release the straw.

20 Claims, 4 Drawing Sheets



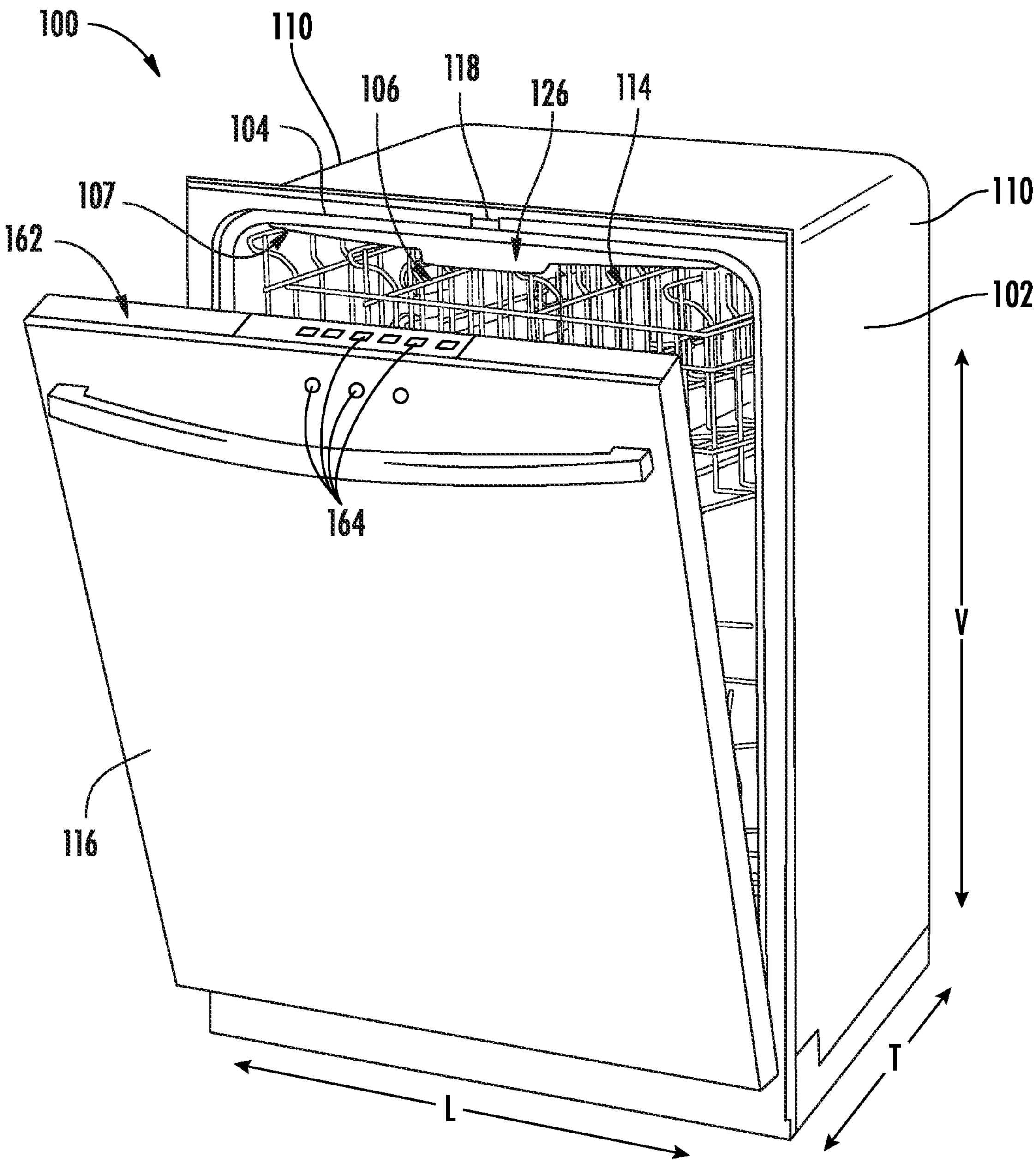


FIG. 1

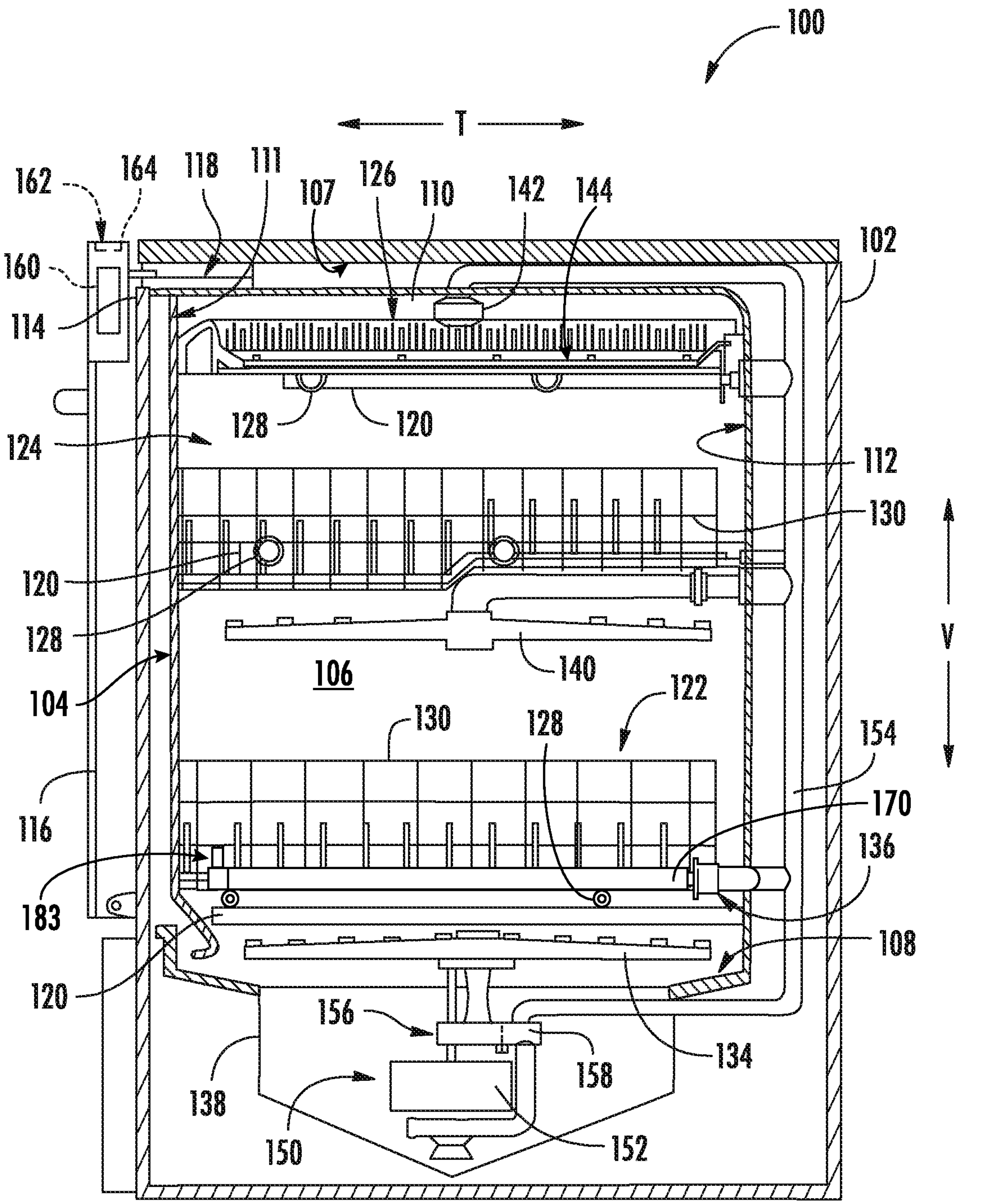

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FIG. 2

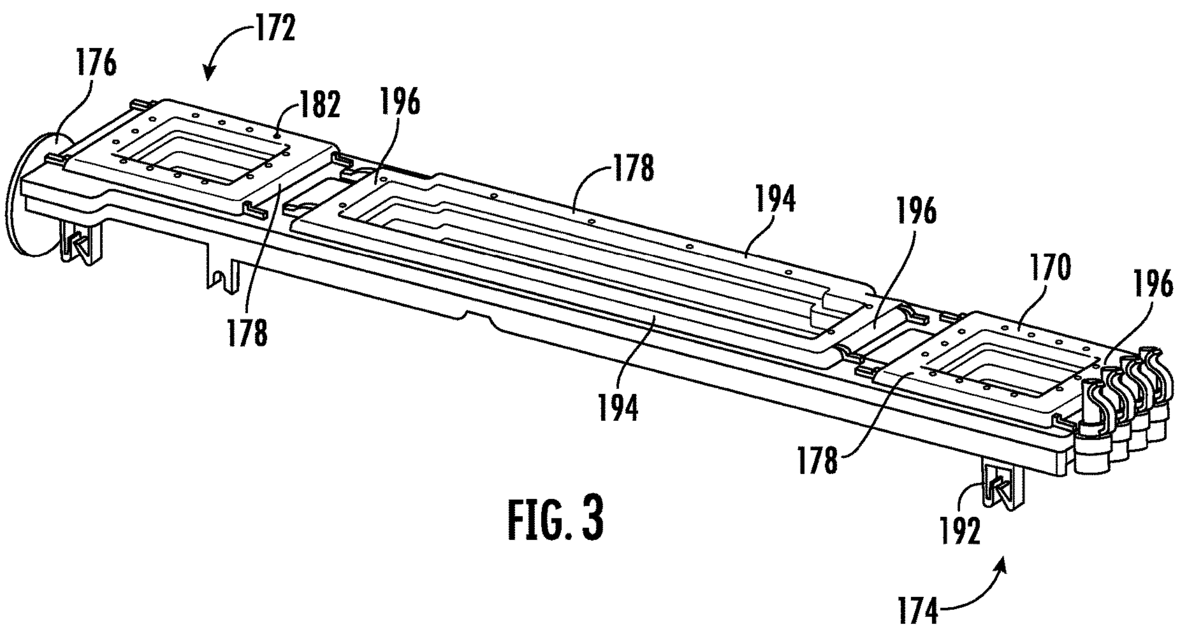
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FIG. 3

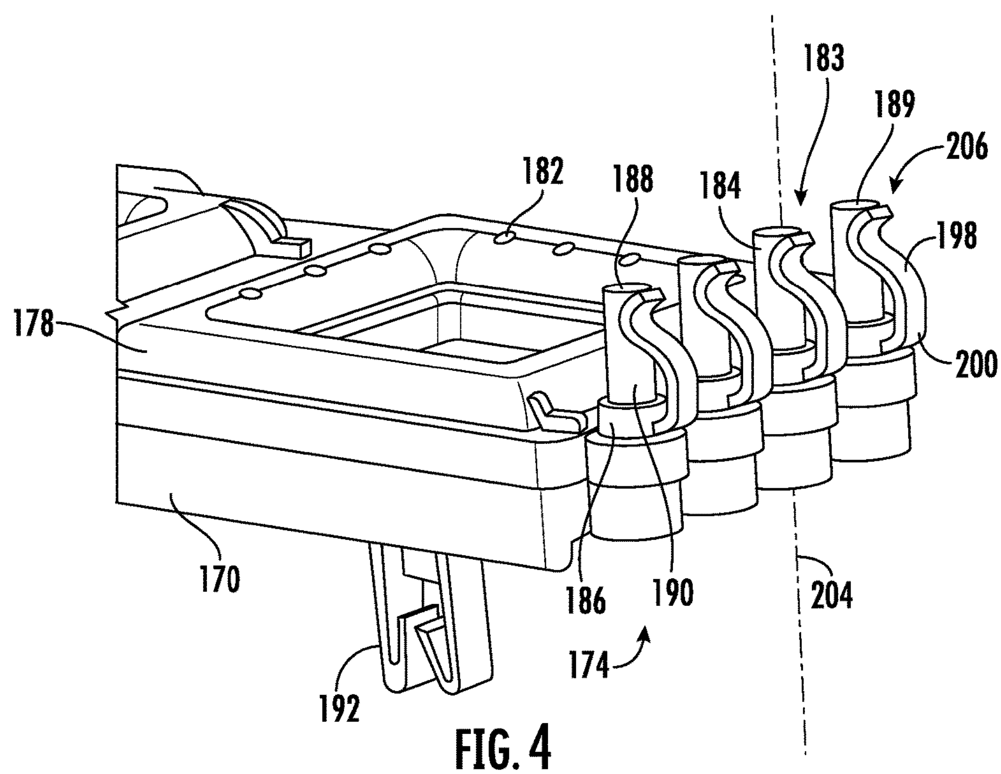
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FIG. 4

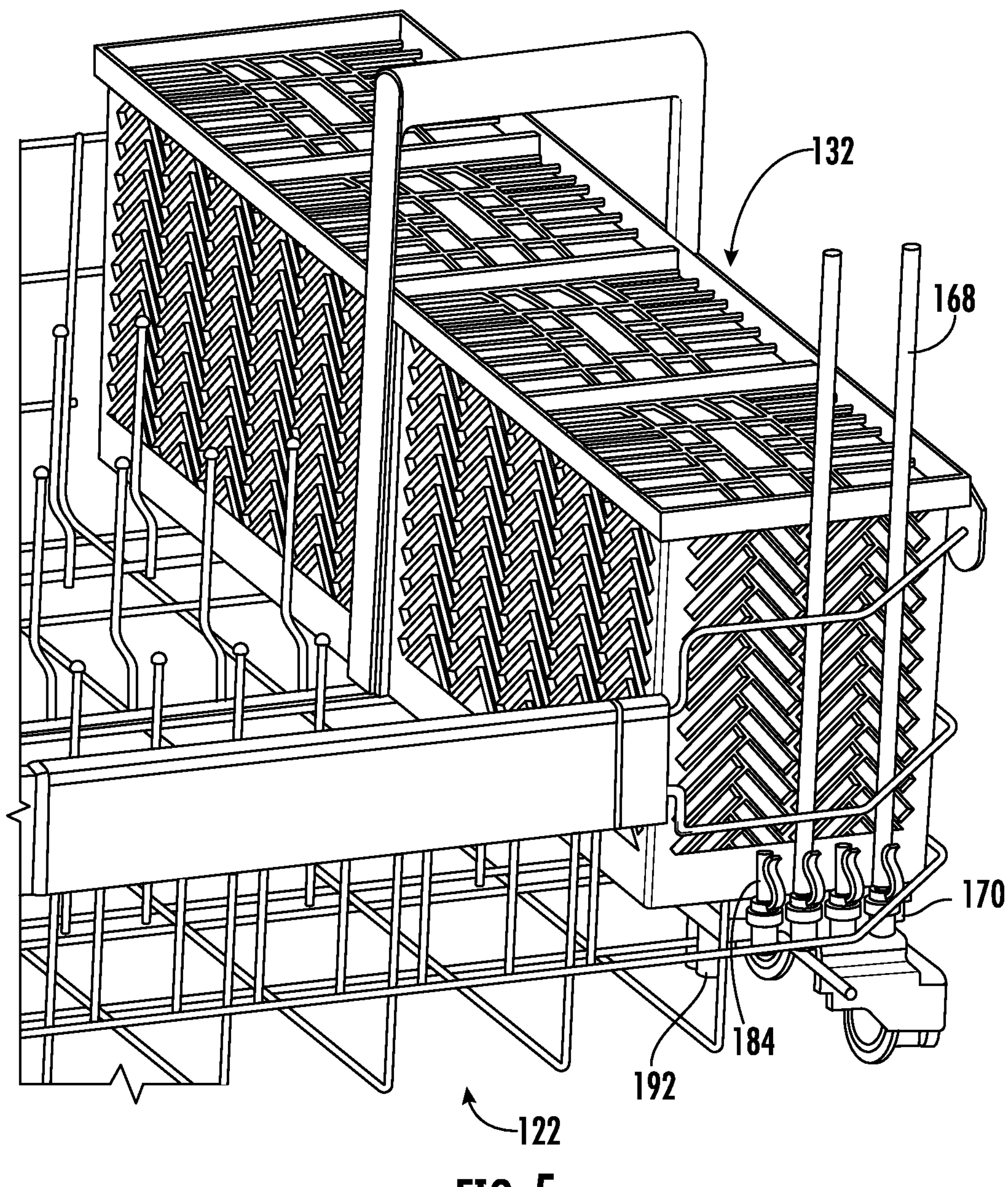

FIG. 5

STRAW HOLDER ASSEMBLY FOR A DISHWASHER APPLIANCE

FIELD OF THE INVENTION

The present disclosure relates generally to dishwasher appliances, and more particularly to a washing system for straws within a dishwasher.

BACKGROUND OF THE INVENTION

Dishwasher appliances generally include a tub that defines a wash chamber and a rack assembly for the receipt of articles for washing. Some dishwasher appliances include a basket for silverware and similar items, with the basket attached to a rack assembly. Generally, the rack assembly and the silverware basket can accommodate most articles selected for washing in a dishwasher.

Recently, durable, reusable straws have become popular and are often washed in a dishwasher. However, the relatively long, tubular configuration of the reusable straws do not lend themselves to adequate cleaning in either a rack or the silverware basket. Typical racks cannot secure the straws in position to sufficiently clean the inner passage. The perforations in the bottom of typical silverware baskets are too large to prevent many straws from passing through the bottom of the basket and interfering with operation of the appliance. Perforations that may sufficiently contain the straws generally do not provide reliable or consistent wash water contact with the inner passage to properly clean the straws.

Accordingly, a dishwasher appliance including a straw holder that addresses the above-mentioned issues would be beneficial.

BRIEF DESCRIPTION OF THE INVENTION

Aspects and advantages of the invention will be set forth in part in the following description, may be apparent from the description, or may be learned through practice of the invention.

In one exemplary aspect, a dishwasher appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular is presented. The dishwasher appliance comprises a tub defining a wash chamber, a rack mounted within the wash chamber and configured for receipt of articles for cleaning, and a wash assembly for retaining a tubular article having an inner passage and introducing a stream of wash fluid to said inner passage. The wash assembly comprises a base defining a conduit, the conduit having a first end and a second end, a nozzle fluidly coupled to the second end of the conduit, the nozzle configured to be received within the inner passage, and a retainer system to removably secure the tubular article to the wash assembly, and wherein the conduit provides a pressurized flow of a wash fluid to the nozzle.

In another exemplary aspect, a wash assembly for retaining a tubular article having an inner passage and introducing a stream of wash fluid to said inner passage is presented. The wash assembly comprises a base defining a conduit, the conduit having a first end and a second end, a nozzle fluidly coupled to the second end of the conduit, the nozzle configured to be received within the inner passage, and a retainer system to removably secure the tubular article to the wash assembly, and wherein the conduit is adapted to provide a pressurized flow of a wash fluid to the nozzle.

These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

FIG. **1** provides a perspective view of an exemplary embodiment of a dishwashing appliance of the present disclosure with a door in a partially open position;

FIG. **2** provides a side, cross-sectional view of the exemplary dishwashing appliance of FIG. **1**;

FIG. **3** provides a perspective view of a base in accordance with an embodiment of the present disclosure;

FIG. **4** provides an enlarged view of the second end of the base of FIG. **3**; and

FIG. **5** provides a perspective view of a rack assembly with an attached wash assembly in accordance with an embodiment of the present invention.

Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

As used herein, the terms "first," "second," and "third" may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms "includes" and "including" are intended to be inclusive in a manner similar to the term "comprising." Similarly, the term "or" is generally intended to be inclusive (i.e., "A or B" is intended to mean "A or B or both"). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as "generally," "about," "approximately," and "substantially," are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., "generally vertical" includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." In addition, references to "an embodiment" or "one embodiment" does not necessarily refer to the same embodiment, although it may. Any implementation described herein as "exemplary" or "an embodiment" is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

Turning to the figures, FIGS. **1** and **2** depict an exemplary domestic dishwasher or dishwashing appliance **100** that may be configured in accordance with aspects of the present disclosure. For the particular embodiment of FIGS. **1** and **2**, the dishwasher **100** includes a cabinet **102** (FIG. **2**) having a tub **104** therein that defines a wash chamber **106**. As shown in FIG. **2**, tub **104** extends between a top **107** and a bottom **108** along a vertical direction V, between a pair of side walls **110** along a lateral direction L, and between a front side **111** and a rear side **112** along a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular to one another.

The tub **104** includes a front opening **114** and a door **116** hinged at its bottom for movement between a normally closed vertical position (shown in FIG. **2**), wherein the wash chamber **106** is sealed shut for washing operation, and a horizontal open position for loading and unloading of articles from the dishwasher **100**. According to exemplary embodiments, dishwasher **100** further includes a door closure mechanism or assembly **118** that is used to lock and unlock door **116** for accessing and sealing wash chamber **106**.

As best illustrated in FIG. **2**, tub side walls **110** accommodate one or more rack assemblies (three shown). More specifically, guide rails **120** may be mounted to side walls **110** for supporting a lower rack assembly **122**, a middle rack assembly **124**, and an upper rack assembly **126** for sliding motion. As illustrated, upper rack assembly **126** is positioned at a top portion of wash chamber **106** above middle rack assembly **124**, which is positioned above lower rack assembly **122** along the vertical direction V. Each rack assembly **122**, **124**, **126** is adapted for sliding movement on the guide rails **120** between an extended loading position (second position, not shown) in which the rack is positioned at least partially outside the wash chamber **106**, and a retracted position (first position as shown in FIGS. **1** and **2**) in which the rack is located inside the wash chamber **106**. This is facilitated, for example, by rollers **128** mounted onto rack assemblies **122**, **124**, **126**, respectively. Although guide rails **120** and rollers **128** are illustrated herein as facilitating movement of the respective rack assemblies **122**, **124**, **126**, it should be appreciated that any suitable sliding mechanism or member may be used according to alternative embodiments.

Some, or all, of the rack assemblies **122**, **124**, **126** are fabricated into lattice structures including a plurality of wires or elongated members **130** (for clarity of illustration, not all elongated members making up rack assemblies **122**, **124**, **126** are shown in FIG. **2**). In this regard, rack assemblies **122**, **124**, **126** are generally configured for supporting articles within wash chamber **106** while allowing a flow of wash fluid to reach and impinge on those articles, e.g., during a cleaning or rinsing cycle. According to another exemplary embodiment, a silverware basket (not shown) may be removably attached to a rack assembly, e.g., lower rack assembly **122**, for placement of silverware, utensils, and the like, that are otherwise too small to be accommodated by rack **122**.

Dishwasher **100** further includes a plurality of spray assemblies for urging a flow of water or wash fluid onto the articles placed within wash chamber **106**. More specifically, as illustrated in FIG. **2**, dishwasher **100** includes a lower spray arm assembly **134** disposed in a lower region **136** of wash chamber **106** and above a sump **138** so as to rotate in relatively close proximity to lower rack assembly **122**. Similarly, a mid-level spray arm assembly **140** is located in an upper region of wash chamber **106** and may be located below and in close proximity to middle rack assembly **124**. In this regard, mid-level spray arm assembly **140** may generally be configured for urging a flow of wash fluid up through middle rack assembly **124** and upper rack assembly **126**. Additionally, an upper spray assembly **142** may be located above upper rack assembly **126** along the vertical direction V. In this manner, upper spray assembly **142** may be configured for urging and/or cascading a flow of wash fluid downward over rack assemblies **122**, **124**, and **126**. As further illustrated in FIG. **2**, upper rack assembly **126** may further define an integral spray manifold **144**, which is generally configured for urging a flow of wash fluid substantially upward along the vertical direction V through upper rack assembly **126**.

The various spray assemblies and manifolds described herein may be part of a fluid distribution system or fluid circulation assembly **150** for circulating water and wash fluid in the tub **104**. More specifically, fluid circulation assembly **150** includes a pump **152** for circulating water and wash fluid (e.g., detergent, water, and/or rinse aid) in the tub **104**. Pump **152** may be located within sump **138** or within a machinery compartment located below sump **138** of tub **104**, as generally recognized in the art. Fluid circulation assembly **150** may include one or more fluid conduits or circulation piping for directing water and/or wash fluid from pump **152** to the various spray assemblies and manifolds. For example, as illustrated in FIG. **2**, a primary supply conduit **154** may extend from pump **152**, along rear **112** of tub **104** along the vertical direction V to supply wash fluid throughout wash chamber **106**.

As illustrated, primary supply conduit **154** is used to supply pressurized wash fluid to one or more spray assemblies, e.g., to mid-level spray arm assembly **140** and upper spray assembly **142**. However, it should be appreciated that according to alternative embodiments, any other suitable plumbing configuration may be used to supply pressurized wash fluid throughout the various spray manifolds and assemblies described herein. For example, according to another exemplary embodiment, primary supply conduit **154** could be used to provide pressurized wash fluid to mid-level spray arm assembly **140** and a dedicated secondary supply conduit (not shown) could be utilized to provide pressurized wash fluid to upper spray assembly **142**. Other plumbing configurations may be used for providing pressurized wash fluid to the various spray devices and manifolds at any location within dishwasher appliance **100**.

According to an embodiment, primary supply conduit **154** may also supply a pressurized flow of wash fluid to base **170** (FIG. **2**), the base **170** affixed to a rack, as illustrated lower rack assembly **122**. The base **170** defines a network of one or more passageways **178** formed internally to the base **170** and generally extending from first end **172** to second end **174**. The primary supply conduit **154** may be in fluid communication with a first coupling **176** positioned at the first end **172** when the rack assembly **122** is fully received within the wash chamber **106**. Primary supply conduit **154** may include a second coupling **180** configured to fluidly join with first coupling **176**. The first and second couplings **176**, **180** may provide a separable fluid connection to support the rack **122** sliding out of the wash chamber **106** (i.e., second position) and reestablishing the fluid connection when the rack **122** is returned to the first position. Accordingly, the first end **172** of the passageway **178** is in fluid communication with the source of the pressurized flow of wash fluid at least when the rack **122** is in the first position.

Each spray arm assembly **134**, **140**, **142**, integral spray manifold **144**, or other spray device may include an arrangement of discharge ports or orifices for directing wash fluid received from pump **152** onto dishes or other articles located in wash chamber **106**. The arrangement of the discharge ports, also referred to as jets, apertures, or orifices, may provide a rotational force by virtue of pressurized wash fluid flowing through the discharge ports. Alternatively, spray arm assemblies **134**, **140**, **142** may be motor-driven, or may operate using any other suitable drive mechanism. Spray manifolds and assemblies may also be stationary. The resultant movement of the spray arm assemblies **134**, **140**, **142** and the spray from fixed manifolds provides coverage of dishes and other dishwasher contents with a washing spray. Other configurations of spray assemblies may be used as well. For example, dishwasher **100** may have additional spray assemblies for cleaning silverware, for scouring casserole dishes, for spraying pots and pans, for cleaning bottles, etc.

As illustrated in FIG. **5**, an optional silverware basket **132** may be supported by a portion of base **170** which may provide a flow of pressurized wash fluid through orifices **182** (FIGS. **3** and **4**). According to another embodiment, dedicated wash assemblies **183** with nozzles **184** may be provided at the second end **174** of base **170** for cleaning tubular articles **168**, such as reusable straws. In particular, the wash assemblies **183** retain the tubular article **168** having an inner passage and introduce a stream of wash fluid to the inner passage. The nozzles **184** are in fluid communication with the passageways **178** and receive the flow of pressurized wash fluid from the pump **152**. Additionally or alternatively, other spray assemblies may be provided. One skilled in the art will appreciate that the embodiments discussed herein are used for the purpose of explanation only and are not limitations of the present subject matter.

In operation, pump **152** draws wash fluid in from sump **138**, pressurizes the fluid, and pumps it to a diverter assembly **156**, e.g., which is positioned within sump **138** of dishwasher appliance. Diverter assembly **156** may include a diverter disk (not shown) disposed within a diverter chamber **158** for selectively distributing the pressurized wash fluid to the spray arm assemblies **134**, **140**, **142** and/or other spray manifolds or devices. For example, the diverter disk may have a plurality of apertures that are configured to align with one or more outlet ports (not shown) at the top of diverter chamber **158**. In this manner, the diverter disk may be selectively rotated to provide wash fluid to the desired spray device.

According to an exemplary embodiment, diverter assembly **156** is configured for selectively distributing the flow of pressurized wash fluid from pump **152** to various fluid supply conduits, only some of which are illustrated in FIG. **2** for clarity. More specifically, diverter assembly **156** may include four outlet ports (not shown) for supplying pressurized wash fluid to a first conduit for rotating lower spray arm assembly **134**, a second conduit for rotating mid-level spray arm assembly **140**, a third conduit for spraying upper spray assembly **142**, and the fourth to a network of passageways **178** to supply orifices **182** and nozzles **184** with pressurized wash fluid. According to some embodiments, the pump **152** and the supply conduits directing the pressurized flow of wash fluid to the nozzles **184** produce a stream of wash fluid at the nozzle extending about 8 to 16 inches from the nozzle outlet **189**, or about 10 to 14 inches from the nozzle outlet, or about 12 inches from the nozzle outlet **189**. The nozzle **184** directs the pressurized flow of wash fluid generally parallel to nozzle axis **204** (FIG. **4**). In use, the nozzle introduces a stream of wash fluid to an inner passage of the tubular article **168**.

The dishwasher **100** is further equipped with a controller **160** to regulate operation of the dishwasher **100**. The controller **160** may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. Alternatively, controller **160** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software.

The controller **160** may be positioned in a variety of locations throughout dishwasher **100**. In the illustrated embodiment, the controller **160** may be located within a control panel area **162** of door **116** as shown in FIGS. **1** and **2**. In such an embodiment, input/output ("I/O") signals may be routed between the control system and various operational components of dishwasher **100** along wiring harnesses that may be routed through the bottom of door **116**. Typically, the controller **160** includes a user interface panel/controls **164** through which a user may select various operational features and modes and monitor progress of the dishwasher **100**. In one embodiment, the user interface **164** may represent a general purpose I/O ("GPIO") device or functional block. In one embodiment, the user interface **164** may include input components, such as one or more of a variety of electrical, mechanical or electro-mechanical input devices including rotary dials, push buttons, and touch pads.

The user interface **164** may include a display component, such as a digital or analog display device designed to provide operational feedback to a user. The user interface **164** may be in communication with the controller **160** via one or more signal lines or shared communication busses.

It should be appreciated that the invention is not limited to any particular style, model, or configuration of dishwasher **100**. The exemplary embodiment depicted in FIGS. **1** and **2** is for illustrative purposes only. For example, different locations may be provided for user interface **164**, different configurations may be provided for rack assemblies **122**, **124**, **126**, different spray arm assemblies **134**, **140**, **142** and spray manifold configurations may be used, and other differences may be applied while remaining within the scope of the present subject matter.

In the exemplary example illustrated in FIG. **3**, base **170** is a generally rectangular structure defining fluid passageways **178** arranged in an interconnected network comprising pairs of elongate rails **194**, generally parallel to each other, connected by cross pieces **196**. Other embodiments include other configurations of defined fluid passageways. One or more attachment devices or clips **192** may be provided to secure the base **170** to the rack assembly **122** such that the nozzle axis **204** is generally parallel with the vertical direction V. As illustrated, the clips **192** are provided on the bottom surface of the base **170**. In other embodiments, the clips **192** may be positioned at other locations on the base **170**, for example on the sides corresponding to the rails **194** or cross piece **196**. The clips **192** are configured to removably affix the base **170** to a rack assembly, for example lower rack assembly **122**. Clips **192** may be provided to affix the base to the middle and upper rack assemblies **124**, **126**.

Returning to FIG. **4**, wash assemblies **183**, including nozzles **184** are shown at second end **174** of base **170** for ease of illustration. Additionally or alternatively, nozzles **184** may be located at first end **172**, or at any other portion of the base **170** in fluid communication with passageways **178**. Four nozzles **184** are illustrated for convenience. In other embodiments more or fewer nozzles may be included. Nozzles **184** may be formed with base **170** or may be separately formed and attached to the base and fluidly coupled to the passageways. The nozzles may be attached to the base either individually or as a multi-nozzle block or manifold.

As mentioned above, the nozzles **184** may be provided as a dedicated source of pressurized wash fluid for cleaning tubular articles **168**, such as durable, reusable straws, particularly the inside surface or such articles. As such, the illustrative nozzles **184** comprise a proximal end **186** adjacent to the base **170** and a distal end **188** spaced from the proximal end **186**. The distal end **188** includes an orifice providing a nozzle outlet **189**. In the illustrative embodiment of FIG. **4**, the nozzle **184** has a generally tubular body, body **190** extending from the proximal end **186** to the distal end **188** and configured to be received within an inner passage, for example the inner passage of a tubular body or straw. In general, the body **190** may be formed from a durable material which may be pliant, for example comprising a natural or synthetic elastomer or elastomeric compound, or rigid, for example a plastic or metal.

In some embodiments, the body **190** may be tapered from the distal end **188** to the proximal end **186**. In other words, the outer surface of the body **190** may be tapered radially outward from the free or distal end **188** to the proximal end **186**, proximate to the base **170**. The taper may be gradual or stepped such that a plurality of tubular articles **168** of different internal dimensions may fit on the body **190**. The internal dimension may be a diameter or may be other internal dimensions depending on the tubular configuration (e.g., oval or rectangular).

According to some embodiments, a retainer system **206** may be provided to removably secure the tubular article **168** to the wash assembly **183**, for example to resist the force of the pressurize wash fluid directed to the inside surfaces of the tubular articles **168**. The retainer system **206** may comprise the nozzle **184** alone, the body **190** sized and shaped to cooperate with one or more sizes of tubular articles **168**, or the retainer system **206** may include additional components.

In some embodiments of a retainer system **206**, at least the outer surface of the body **190** comprises an elastomer that may deform when the nozzle body **190** is received in the tubular article **168** (i.e., the tubular article is placed over the nozzle **184**). As such, the body **190** may engage the tubular article **168** to provide support and may form a seal with the inner wall of the tubular article **168**. As discussed above, a portion of the body **190** may be radially tapered, increasing in diameter from the distal end **188** to the proximal end **186**. As such, the body **190** may engage and support smaller diameter tubular articles **168** at the distal end **188** as part of a retainer system **206**. Progressively large tubular articles **168** may engage and be retained by portions of the body **190** lower on the body **190** (i.e. closer to the proximal end **186**). In other embodiments, the tubular article **168** may fit loosely on the body **190** and may utilize additional components to secure the tubular article **168** to the wash assembly **183**.

In other embodiments of a retainer system **206**, wash assembly **183** may include a retention clip **198** provided to removably secure the tubular article **168** to the nozzle **184**. In the illustrative embodiment of FIGS. **4** and **5**, retention clip **198** comprises a first or proximal end **200** secured to the nozzle **184** against displacement with respect to the nozzle **184**. The retention clip **198** further comprises a second end **202** resiliently deformable from a first position in which the second end **202** is proximate to the distal end **188** of the body **190** and a second position in which the second end is spaced apart from the distal end **188** of the body **190**. In the second position, the second end **202** is resiliently deformed in a generally radial direction from the first position. Resiliently deforming the second end **202** from the first position to the second position forms a variable gap between the second end **202** and the nozzle body **190**. In use, the second end **202** of the retention clip **198** is resiliently deformed away from the body **190** to allow the tubular article **168** to be disposed over the body **190**. The second end **202** of the retention clip **198** returns to its original position to removably secure the tubular article **168** to the body **190**.

The above-described retainer systems **206** are examples of suitable systems for retaining a tubular article **168** on the wash assembly **183**, provided as examples and not limitations. Other retainer systems may be used within the scope of the present disclosure.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. A dishwasher appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the dishwasher appliance comprising:
- a tub defining a wash chamber;
- a rack mounted within the wash chamber and configured for receipt of articles for cleaning; and
- a wash assembly for retaining a tubular article having an inner passage and introducing a stream of wash fluid to said inner passage, the wash assembly comprising:
  - a base defining a conduit, the conduit having a first end and a second end;
  - a nozzle fluidly coupled to the second end of the conduit, the nozzle configured to be received within the inner passage; and
  - a retainer system to removably secure the tubular article to the wash assembly; and
- wherein the conduit provides a pressurized flow of a wash fluid to the nozzle.

2. The dishwasher appliance of claim 1, wherein the rack is supported for sliding motion between a first position in which the rack is fully retracted within the wash chamber and a second position in which the rack extends at least partially from the wash chamber.

3. The dishwasher appliance of claim 2, wherein:
- the nozzle defines a nozzle axis; and
- the base is affixed to the rack such that the nozzle axis is generally parallel to the vertical direction.

4. The dishwasher appliance of claim 3, wherein the first end of the conduit is in fluid communication with a source of the pressurized flow of wash fluid at least when the rack is in the first position.

5. The dishwasher appliance of claim 4, wherein a pump provides the pressurized flow to the nozzle and the nozzle directs the pressurized flow generally parallel to the nozzle axis.

6. The dishwasher appliance of claim 5, wherein the pressurized flow produces a stream of wash fluid extending about 12 inches from a nozzle outlet.

7. The dishwasher appliance of claim 1, wherein:
- the nozzle comprises a proximal end adjacent to the base and a distal end spaced from the proximal end; and
- the retainer system includes a retention clip at the nozzle, the retention clip configured to removably secure the tubular article to the nozzle.

8. The dishwasher appliance of claim 7, wherein the retention clip comprises:
- a first end secured against displacement with respect to the nozzle; and
- a second end resiliently deformable from a first position proximate to the distal end and a second position spaced apart from the distal end.

9. The dishwasher appliance of claim 8, wherein, in the second position, the second end is resiliently deformed in a generally radial direction from the first position.

10. The dishwasher appliance of claim 9, wherein resiliently deforming the second end from the first position to the second position forms a variable gap between the second end and the nozzle.

11. The dishwasher appliance of claim 1, further comprising a plurality of nozzles fluidly coupled to the conduit.

12. The dishwasher appliance of claim 1, wherein the nozzle comprises a proximal end adjacent to the base, a distal end spaced from the proximal end, and an outer surface, the outer surface tapering radially outward from the distal end to the proximal end.

13. A wash assembly for retaining a tubular article having an inner passage and introducing a stream of wash fluid to said inner passage, the wash assembly comprising:
- a base defining a conduit, the conduit having a first end and a second end;
- a nozzle fluidly coupled to the second end of the conduit, the nozzle configured to be received within the inner passage; and
- a retainer system to removably secure the tubular article to the wash assembly; and
- wherein the conduit is adapted to provide a pressurized flow of a wash fluid to the nozzle.

14. The wash assembly of claim 13, wherein:
- the nozzle comprises a proximal end adjacent to the base and a distal end spaced from the proximal end; and
- the retainer system further comprises a retention clip at the nozzle, the retention clip configured to removably secure the tubular article to the nozzle.

15. The wash assembly of claim 14, wherein the retention clip comprises:
- a first end secured against displacement with respect to the nozzle; and
- a second end resiliently deformable from a first position proximate to the distal end and a second position spaced apart from the distal end.

16. The wash assembly of claim 15, wherein, in the second position, the second end is resiliently deformed in a generally radial direction from the first position.

17. The wash assembly of claim 16, wherein resiliently deforming the second end from the first position to the second position forms a variable gap between the second end and the nozzle.

18. The wash assembly of claim 13, further comprising a plurality of nozzles fluidly coupled to the conduit.

19. The wash assembly of claim 13, wherein the nozzle comprises a proximal end adjacent to the base, a distal end spaced from the proximal end, and an outer surface, the outer surface tapering radially outward from the distal end to the proximal end.

20. The wash assembly of claim 19, wherein the outer surface comprises an elastomer.

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